

Introduction to Frequency-Response Analysis

What a system does to a sinusoid: gain and phase vs. frequency.

Ogata, *Modern Control Engineering*, Ch. 7.

In this tutorial you will:

- see why the steady-state sinusoidal response is captured by $G(j\omega)$,
- verify the gain/phase prediction against a simulation, and
- meet the Bode, polar, and Nichols ways of plotting $G(j\omega)$.

Step through with **Ctrl+Enter**, or render a report with **publish**.

The frequency response of a system is its steady-state response to a sinusoidal input. For a stable, linear time-invariant system with transfer function $G(s)$, if the input is

$$u(t) = A \sin(\omega t)$$

then the steady-state output is also sinusoidal, at the same frequency, but scaled in magnitude and shifted in phase:

$$y(t) = AM(\omega) \sin(\omega t + \phi(\omega))$$

where $M(\omega) = |G(j\omega)|$ and $\phi(\omega) = \angle G(j\omega)$. This follows directly from substituting $s = j\omega$ into the transfer function – $G(j\omega)$ is called the **sinusoidal transfer function**.

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Why $s = j\omega$

The Laplace transform of $A \sin(\omega t)$ is $\frac{A\omega}{s^2 + \omega^2}$, with poles at $s = \pm j\omega$ on the imaginary axis. For a stable $G(s)$, the partial-fraction expansion of $Y(s) = G(s)U(s)$ separates into terms from $G(s)$'s own poles (which decay to zero, the transient) and a term from the input poles at $\pm j\omega$, which persists forever as the steady-state sinusoidal response. Evaluating the residues at $s = \pm j\omega$ shows the steady-state output is fully characterized by $G(j\omega)$.

Worked Example

$G(s) = \frac{1}{s+1}$. Evaluate $G(j\omega)$ at a few frequencies and confirm against `bode/|freqresp|`.

```
G = tf(1,[1 1]);
omegas = [0.1 1 2 10];
for w = omegas
    g = evalfr(G,1j*w);
    fprintf('w=%5.2f: |G(jw)|=%.4f, angle=%.2f deg\n', w, abs(g), angle(g)*180/pi)
end
```

```
w= 0.10: |G(jw)|=0.9950, angle=-5.71 deg
w= 1.00: |G(jw)|=0.7071, angle=-45.00 deg
w= 2.00: |G(jw)|=0.4472, angle=-63.43 deg
w=10.00: |G(jw)|=0.0995, angle=-84.29 deg
```

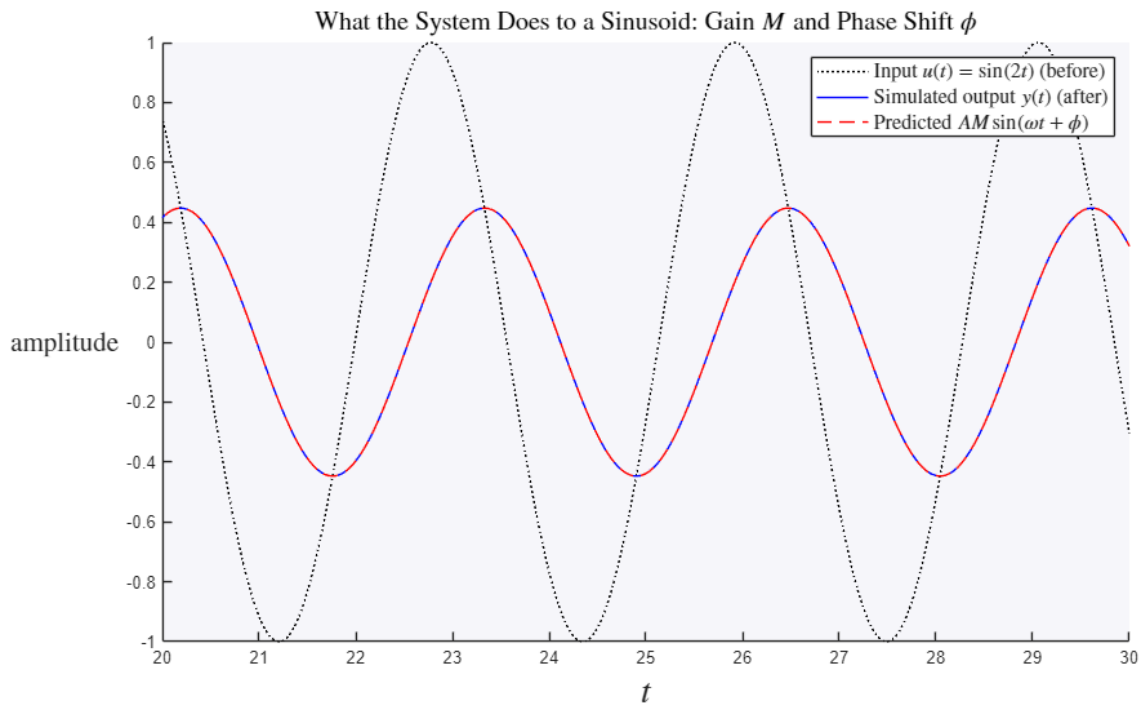
Direct Verification via Simulation

Drive $G(s)$ with $u(t) = \sin(2t)$ and confirm the steady-state output amplitude/phase match $|G(j2)|, \angle G(j2)$.

```
w_test = 2;
t = 0:0.01:30;
u = sin(w_test*t);
y = lsim(G,u,t);

g_test = evalfr(G,1j*w_test);
M = abs(g_test); phi = angle(g_test);
y_pred = M*sin(w_test*t + phi);

figure
hold on
plot(t,u,'k:')
plot(t,y,'b')
plot(t,y_pred,'r--')
hold off
legend('Input  $u(t)=\sin(2t)$  (before)', 'Simulated output  $y(t)$  (after)', ...
       'Predicted  $AM\sin(\omega t + \phi)$ ', 'Interpreter','latex','FontSize',12)
title('What the System Does to a Sinusoid: Gain  $M$  and Phase Shift  $\phi$ ', 'Interpreter','latex','FontSize',15)
ylabel('amplitude', 'Interpreter','latex','FontSize',16)
set(gca, 'YLabel', 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)
xlim([20 30])
```



Read the "before vs. after" directly off the plot: the output is the input scaled by $M = |G(j2)|$ and delayed in phase by $\phi = \angle G(j2)$.

```
fprintf('At w=2: gain M = %.4f (output/input amplitude), phase = %.2f deg\n', M, phi*180/pi)
```

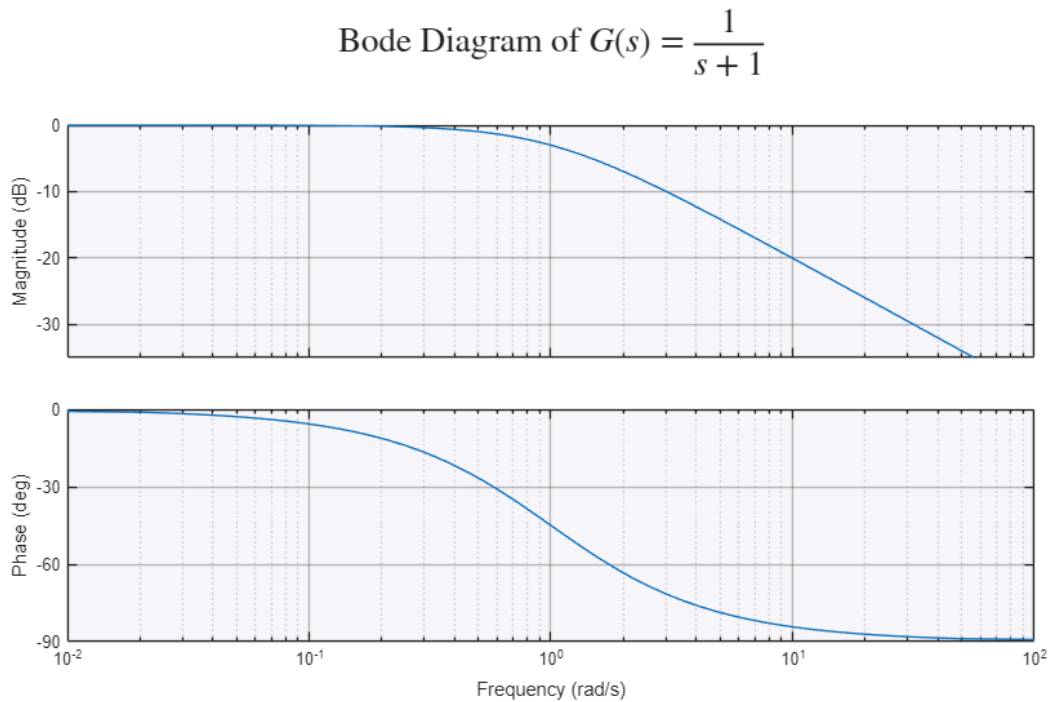
```
At w=2: gain M = 0.4472 (output/input amplitude), phase = -63.43 deg
```

Standard Frequency-Response Plots

Three equivalent graphical representations of $G(j\omega)$ are used throughout this chapter and the next:

1. **Bode diagram:** $20 \log_{10} |G(j\omega)|$ (dB) and $\angle G(j\omega)$ (deg), each vs. $\log_{10} \omega$.
2. **Polar (Nyquist) plot:** $G(j\omega)$ traced in the complex plane as ω sweeps $0 \rightarrow \infty$.
3. **Log-magnitude-vs-phase (Nichols) plot:** magnitude (dB) vs. phase (deg), parameterized by ω .

```
figure
bode(G)
title('Bode Diagram of $G(s)=\frac{1}{s+1}$','Interpreter','latex','FontSize',20)
grid on
```



Try it yourself

- Change the test frequency `w_test` to 0.5 and 5 and notice the output amplitude follow $|G(j\omega)|$ (bigger at low frequency for this lag).
- Read the phase lag off the shifted output and compare with `angle(evalfr(G,1j*w_test))`.